

# NETWORKING 09

## BASIC FIREWALL CONCEPT

### What is a Firewall?

Think of a firewall like a **security guard at your house gate** 

- It checks **who is coming in (inbound)**
- It checks **what is going out (outbound)**
- It allows or blocks based on **rules**

👉 In simple words:

**Firewall = filter for internet traffic**

### **Firewall Controls**

#### **Inbound Traffic (Coming INSIDE)**

- Data coming **from internet** → **your laptop/server**
- Example:
  - Someone trying to open your website
  - Hacker trying to access your system ❌

👉 Firewall checks:

"Should I allow this person or block?"

#### **Outbound Traffic (Going OUTSIDE)**

- Data going **from your system** → **internet**
- Example:
  - You opening Google
  - App sending data online

👉 Firewall checks:

"Is this safe to send or suspicious?"

# Rules (Very Important )

Firewall works based on **rules** (like instructions)

## **Example 1: Allow HTTP**

- HTTP = normal web browsing (port 80)
- Rule:  
    👉 "Allow users to open websites"

✓ So you can use browser normally

## **Example 2: Block Unknown Ports**

- Ports = doors of your computer
- Unknown ports = suspicious doors 

👉 Rule:

"Block anything coming from unknown/untrusted ports"

✓ Stops hackers entering through hidden ways

👉 **Firewall does NOT only check ports, it checks:**

- Port number 
- IP address 
- Protocol (HTTP, HTTPS, FTP, etc.)
- Sometimes even data (advanced firewalls)

👉 **But for beginner level:**

✓ **Ports = main thing firewall uses to allow/block**

# Your Statement (Corrected)

You said:

"If any website uses unknown port firewall blocks it"

  **PARTLY CORRECT**

## **Correct explanation:**

- If a request comes from a **port not allowed in firewall rules** →  Blocked
- If port is allowed →  Allowed

 So firewall follows rules like:

- Allow port 80 (HTTP)
- Allow port 443 (HTTPS)
- Block all others

## **Important Ports (Must Remember)**

### **HTTP (Normal website)**

- Port = **80**

### **HTTPS (Secure website)**

- Port = **443**

 So answer to your question:

 **HTTPS uses port 443**

## **Easy Trick to Remember**

- **80** → Open (no security)
- **443** → Secure (HTTPS )